

Complete mitogenomes of *Anopheles peditaeniatus* and *Anopheles nitidus* and phylogenetic relationships within the genus *Anopheles* inferred from mitogenomes.

PMID: 34488869 | Indexed in PubMed

Despite the medical importance of mosquitoes of the genus *Anopheles* in the transmission of malaria and other human diseases, its phylogenetic relationships are not settled, and the characteristics of mitochondrial genome (mitogenome) are not thoroughly understood. The present study sequenced and analyzed the complete mitogenomes of *An. peditaeniatus* and *An. nitidus*, investigated genome characteristics, and inferred the phylogenetic relationships of 76 *Anopheles* spp. The complete mitogenomes of *An. peditaeniatus* and *An. nitidus* are 15,416 and 15,418 bp long, respectively, and both include 13 PCGs, 22 tRNAs, two tRNAs and one control region (CR). Mitogenomes of *Anopheles* spp. are similar to those of other insects in general characteristics; however, the *trnR* and *trnA* have been reversed to "trnR-trnA," as has been reported in other mosquito genera. Genome variations mainly occur in CR length (493-886 bp) with six repeat unit types identified for the first time that demonstrate an evolutionary signal. The subgenera *Lophopodomyia*, *Stethomyia*, *Kerteszia*, *Nyssorhynchus*, *Anopheles* and *Cellia* are inferred to be monophyletic, and the phylogenetic analyses support a new phylogenetic relationship among the six subgenera investigated, in that subgenus *Lophopodomyia* is the sister to all other five subgenera, and the remaining five subgenera are divided into two clades, one of which is a sister-taxon subgenera *Stethomyia* + *Kerteszia*, and the other consists of subgenus *Nyssorhynchus* as the sister to a sister-group subgenera *Anopheles* + *Cellia*. Four series (*Neomyzomyia*, *Pyretophorus*, *Neocellia* and *Myzomyia*) of the subgenus *Cellia*, and two series (*Arribalzagia* and *Myzorhynchus*) of the subgenus *Anopheles* were found to be monophyletic, whereas three sections (*Myzorhynchella*, *Argyritarsis* and *Albimanus*) and their subdivisions of the subgenus *Nyssorhynchus* were polyphyletic or paraphyletic. The study comprehensively uncovered the characteristics of mitogenome and the phylogenetics based on mitogenomes in the genus *Anopheles*, and provided information for further study on the mitogenomes, phylogenetics and taxonomic revision of the genus.

Description of *Anopheles gabonensis*, a new species potentially involved in rodent malaria transmission in Gabon, Central Africa.

PMID: 24840150 | Indexed in PubMed

The genus *Anopheles* includes mosquito vectors of human malaria and arboviruses. In sub-Saharan Africa, the anopheline fauna is rich of nearly 150 species, few of which are anthropophilic and capable of transmitting pathogens to humans. Some of the remaining species are found in forests far from human environments and are vectors of wildlife pathogens. The diversity and the biology of these species have yet to be fully described. As a contribution to furthering knowledge of sylvan Anophelinae, using morphological and molecular tools we describe a new *Anopheles* species collected in Gabon (Central Africa), which we have named *Anopheles gabonensis* n. sp. We also molecularly screened this species to detect infections by *Plasmodium* parasites. The results showed the species to have been infected by *Plasmodium vinckei*, a rodent parasite. We discuss the role of *An. gabonensis* n. sp. in the transmission of *P. vinckei* in the rainforest areas of Central Africa and its potential to transfer pathogens to humans.

Exogenous gypsy insulator sequences modulate transgene expression in the malaria

vector mosquito, *Anopheles stephensi*.

PMID: 23584017 | Indexed in PubMed

Malaria parasites are transmitted to humans by mosquitoes of the genus *Anopheles*, and these insects are the targets of innovative vector control programs. Proposed approaches include the use of genetic strategies based on transgenic mosquitoes to suppress or modify vector populations. Although substantial advances have been made in engineering resistant mosquito strains, limited efforts have been made in refining mosquito transgene expression, in particular attenuating the effects of insertions sites, which can result in variations in phenotypes and impacts on fitness due to the random integration of transposon constructs. A promising strategy to mitigate position effects is the identification of insulator or boundary DNA elements that could be used to isolate transgenes from the effects of their genomic environment. We applied quantitative approaches that show that exogenous insulator-like DNA derived from the *Drosophila melanogaster* gypsy retrotransposon can increase and stabilize transgene expression in transposon-mediated random insertions and recombinase-catalyzed, site-specific integrations in the malaria vector mosquito, *Anopheles stephensi*. These sequences can contribute to precise expression of transgenes in mosquitoes engineered for both basic and applied goals.

Who was the first to visualize the malaria parasite?

PMID: 38600596 | Indexed in PubMed

Human malaria, an ancient tropical disease, is caused by infection with protozoan parasites belonging to the genus *Plasmodium* and is transmitted by female mosquitoes of the genus *Anopheles*. Our understanding of human malaria parasites began officially in 1880 with their discovery in the blood of malaria patients by Charles Louis Alphonse Lavéran (1845-1922), a French army officer working in Algeria. A claim for priority was made by Philipp Friedrich Hermann Klencke (1813-1881) in 1843, who wrote a chapter entitled: "Marvellous parallelism between the manifestations of vertigo and the presence of animalcule vacuoles in living blood." We should not lose sight of this old controversy, which is rarely mentioned in historical reviews on malaria.

Phylogeny of Anophelinae using mitochondrial protein coding genes.

PMID: 29291068 | Indexed in PubMed

Malaria is a vector-borne disease that is a great burden on the poorest and most marginalized communities of the tropical and subtropical world. Approximately 41 species of Anopheline mosquitoes can effectively spread species of *Plasmodium* parasites that cause human malaria. Proposing a natural classification for the subfamily Anophelinae has been a continuous effort, addressed using both morphology and DNA sequence data. The monophyly of the genus *Anopheles*, and phylogenetic placement of the genus *Bironella*, subgenera *Kerteszia*, *Lophopodomyia* and *Stethomyia* within the subfamily Anophelinae, remain in question. To understand the classification of Anophelinae, we inferred the phylogeny of all three genera (*Anopheles*, *Bironella*, *Chagasia*) and major subgenera by analysing the amino acid sequences of the 13 protein coding genes of 150 newly sequenced mitochondrial genomes of Anophelinae and 18 newly sequenced *Culex* species as outgroup taxa, supplemented with 23 mitogenomes from GenBank. Our analyses generally place genus *Bironella* within the genus *Anopheles*,

which implies that the latter as it is currently defined is not monophyletic. With some inconsistencies, Bironella was placed within the major clade that includes Anopheles, Cellia, Kerteszia, Lophopodomyia, Nyssorhynchus and Stethomyia, which were found to be monophyletic groups within Anophelinae. Our findings provided robust evidence for elevating the monophyletic groupings Kerteszia, Lophopodomyia, Nyssorhynchus and Stethomyia to genus level; genus Anopheles to include subgenera Anopheles, Baimaia, Cellia and Christya; Anopheles parvus to be placed into a new genus; Nyssorhynchus to be elevated to genus level; the genus Nyssorhynchus to include subgenera Myzorhynchella and Nyssorhynchus; Anopheles atacamensis and Anopheles pictipennis to be transferred from subgenus Nyssorhynchus to subgenus Myzorhynchella; and subgenus Nyssorhynchus to encompass the remaining species of Argyritarsis and Albimanus Sections.

Opening the toolkit for genetic analysis and control of Anopheles mosquito vectors.

PMID: 30553490 | Indexed in PubMed

Anopheles is the only genus of mosquitoes that transmit human malaria and consequently the focus of large scale genome and transcriptome-wide association studies. Genetic tools to define the function of the candidate genes arising from these analyses are vital. Moreover, genome editing offers the potential to modify Anopheles population structure at local and global scale to provide complementary tools towards the ultimate goal of malaria elimination. Major breakthroughs in Anopheles genetic analysis came with the development of germline transformation and RNA interference technology. Yet, the field has been revolutionised again by precise genome editing now possible through site-specific nucleases. Here we review the components of the current genetic toolkit available to study Anopheles, focusing particularly on how these technical advances are used to gain insight into malaria transmission and the design of genetic methods to control Anopheles vectors.

Malaria parasite development in mosquitoes.

PMID: 9444756 | Indexed in PubMed

Mosquitoes of the genus Anopheles transmit malaria parasites to humans. Anopheles mosquito species vary in their vector potential because of environmental conditions and factors affecting their abundance, blood-feeding behavior, survival, and ability to support malaria parasite development. In the complex life cycle of the parasite in female mosquitoes, a process termed sporogony, mosquitoes acquire gametocyte-stage parasites from blood-feeding on an infected host. The parasites carry out fertilization in the midgut, transform to ookinetes, then oocysts, which produce sporozoites. Sporozoites invade the salivary glands and are transmitted when the mosquito feeds on another host. Most individual mosquitoes that ingest gametocytes do not support development to the sporozoite stage. Bottle-necks occur at every stage of the cycle in the mosquito. Powerful new techniques and approaches exist for evaluating malaria parasite development and for identifying mechanisms regulating malaria parasite-vector interactions. This review focuses on those interactions that are important for the development of new approaches for evaluating and blocking transmission in nature.

orco mutant mosquitoes lose strong preference for humans and are not repelled by volatile DEET.

PMID: 23719379 | Indexed in PubMed

Female mosquitoes of some species are generalists and will blood-feed on a variety of vertebrate hosts, whereas others display marked host preference. *Anopheles gambiae* and *Aedes aegypti* have evolved a strong preference for humans, making them dangerously efficient vectors of malaria and Dengue haemorrhagic fever. Specific host odours probably drive this strong preference because other attractive cues, including body heat and exhaled carbon dioxide (CO₂), are common to all warm-blooded hosts. Insects sense odours via several chemosensory receptor families, including the odorant receptors (ORs), membrane proteins that form heteromeric odour-gated ion channels comprising a variable ligand-selective subunit and an obligate co-receptor called Orco (ref. 6). Here we use zinc-finger nucleases to generate targeted mutations in the orco gene of *A. aegypti* to examine the contribution of Orco and the odorant receptor pathway to mosquito host selection and sensitivity to the insect repellent DEET (N,N-diethyl-meta-toluamide). orco mutant olfactory sensory neurons have greatly reduced spontaneous activity and lack odour-evoked responses. Behaviourally, orco mutant mosquitoes have severely reduced attraction to honey, an odour cue related to floral nectar, and do not respond to human scent in the absence of CO₂. However, in the presence of CO₂, female orco mutant mosquitoes retain strong attraction to both human and animal hosts, but no longer strongly prefer humans. orco mutant females are attracted to human hosts even in the presence of DEET, but are repelled upon contact, indicating that olfactory- and contact-mediated effects of DEET are mechanistically distinct. We conclude that the odorant receptor pathway is crucial for an anthropophilic vector mosquito to discriminate human from non-human hosts and to be effectively repelled by volatile DEET.

Contributions of *Anopheles* larval control to malaria suppression in tropical Africa: review of achievements and potential.

PMID: 17373942 | Indexed in PubMed

Malaria vector control targeting the larval stages of mosquitoes was applied successfully against many species of *Anopheles* (Diptera: Culicidae) in malarious countries until the mid-20th Century. Since the introduction of DDT in the 1940s and the associated development of indoor residual spraying (IRS), which usually has a more powerful impact than larval control on vectorial capacity, the focus of malaria prevention programmes has shifted to the control of adult vectors. In the Afrotropical Region, where malaria is transmitted mainly by *Anopheles funestus* Giles and members of the *Anopheles gambiae* Giles complex, gaps in information on larval ecology and the ability of *An. gambiae* sensu lato to exploit a wide variety of larval habitats have discouraged efforts to develop and implement larval control strategies. Opportunities to complement adulticiding with other components of integrated vector management, along with concerns about insecticide resistance, environmental impacts, rising costs of IRS and logistical constraints, have stimulated renewed interest in larval control of malaria vectors. Techniques include environmental management, involving the temporary or permanent removal of anopheline larval habitats, as well as larviciding with chemical or biological agents. This present review covers large-scale trials of anopheline larval control methods, focusing on field studies in Africa conducted within the past 15 years. Although such studies are limited in number and scope, their results suggest that targeting larvae, particularly in human-made habitats, can significantly reduce malaria transmission in appropriate settings. These approaches are especially suitable for urban areas, where larval habitats are limited,

particularly when applied in conjunction with IRS and other adulticidal measures, such as the use of insecticide treated bednets.

Insecticide susceptibility of *Anopheles coluzzii* and *Anopheles gambiae* mosquitoes in Ibadan, Southwest Nigeria.

PMID: 25417803 | Indexed in PubMed

The emergence of insecticide resistance in *Anopheles* (Diptera: Culicidae) mosquitoes has great implications for malaria control in Nigeria. This study aimed to determine the dynamics of insecticide susceptibility levels and the frequency of knock-down resistance (kdr) mutations (L1014F) in wild *Anopheles coluzzii* Coetzee & Wilkerson sp. n. and *Anopheles gambiae* Giles from the Ojoo and Bodija areas of Ibadan, in southwest Nigeria. Insecticide susceptibility to pyrethroids, organophosphates, carbamates and organochlorines was assessed using World Health Organization (WHO) bioassays. A subset of the mosquitoes exposed to pyrethroids and DDT was used for species and molecular form identification; kdr genotyping was determined using the TaqMan real-time polymerase chain reaction assay. The mosquitoes were resistant to pyrethroids and DDT but completely susceptible to organophosphates and carbamates. Bodija samples (n=186) consisted of *An. gambiae* (91.4%) and *An. coluzzii* (8.1%) and included one *An. coluzzii*/*An. gambiae* hybrid specimen. All mosquitoes screened in Ojoo (n=26) were *An. gambiae*. The 1014F kdr mutation was detected at frequencies of 24.5 and 5.8% in Bodija and Ojoo, respectively. No correlation was observed between kdr genotypes and resistance phenotypes. The results indicate that metabolic resistance probably plays an important role in the development of resistance and highlight the need to implement insecticide resistance management strategies.
